

Ideal Functionalities in the UC Framework: An Encyclopedia in Dependency Order

1 Scope and ordering principle

This document indexes 100 ideal functionalities of the universal composability framework [41, 43]. Entries are ordered by *dependency depth* rather than by importance or by chronology: Layer 0 collects what must be assumed, and each subsequent layer is realizable, at least in principle, from functionalities appearing earlier.

Two conventions require comment.

First, a primitive earns an entry when it exposes a party-facing interface whose ideal behavior can be stated independently of any implementation. Local hardness-based primitives such as one-way functions and pseudorandom generators fail this test: there is no cross-party interface to idealize, and the composition theorem yields nothing that a hybrid argument does not already give. They appear below only so that their absence is recorded explicitly. The partial exception is $\mathcal{F}_{\text{crypto}}$ [109, 7], which is worth having precisely because keys are reused across sessions, that is, only once joint state is present.

Second, several entries in the framework literature are not functionalities but *modifiers* applied to existing ones. These are collected separately in Section 3, because the cross product of modifiers and functionalities generates coverage far more economically than a flat list.

Status markers

- S Idealized setup. Not realizable in the plain model; assumed rather than constructed.
- C Canonical. An established functionality with a widely cited reference definition.
- E Emerging. A UC treatment exists but the formulation is not yet standardized, or competing formulations coexist.
- O Open. No established UC formulation. Writing one is a research contribution, not a citation.

Entries marked O carry no reference. Where a reference is marked with a dagger ($\dagger$), the bibliography entry needs independent verification before use; see the **note** fields in the companion **.bib** file.

2 The index

#	Functionality	Reading	St.	References
Layer 0. Idealized setup and resources				
1	$\mathcal{F}_{\text{CRS}}$	Common reference string	S	[41, 64]
2	$\mathcal{F}_{\text{ACRS}}$	Augmented CRS, “sunspots”	S	[47]
3	$\mathcal{F}_{\text{PKI}}$	Public-key infrastructure	S	[42]
4	$\mathcal{G}_{\text{PKI}}$	Global, non-encapsulated PKI	S	[67]
5	$\mathcal{F}_{\text{REG}}, \mathcal{F}_{\text{KRK}}$	Key registration, with knowledge	S	[17]
6	$\mathcal{F}_{\text{CERT}}$	Certification authority	S	[42]
7	$\mathcal{F}_{\text{RO}}$	Local random oracle	S	[115]

continued on next page

#	Functionality	Reading	St.	References
8	$\mathcal{G}_{\text{RO}}$	Global RO: observable, programmable, restricted	C	[58, 38]
9	$\mathcal{G}_{\text{clock}}$	Clock	S	[100]
10	$\mathcal{F}_{\text{syn}}$	Bounded-delay network	S	[100]
11	$\mathcal{F}_{\text{clocksync}}$	Dynamic ad hoc clock synchronization	C	[11]
12	$\mathcal{F}_{\text{nettime}}$	Attackable network time service	C	[57]
13	$\mathcal{F}_{\text{RP}}$	Ideal cipher, two-sided permutation	S	—
14	$\mathcal{G}_{\text{GG}}$	Generic group, global observable variant; bilinear types 1–3	S	[33]
15	$\mathcal{F}_{\text{wrap}}$	Tamper-proof hardware token	S	[83, 99]
16	$\mathcal{F}_{\text{PUF}}$	Physically uncloneable function	S	[37]
17	$\mathcal{F}_{\text{leak}}$	Leakage, physically observable computation	S	[114, 31]

Layer 1. Channels, agreement, ledgers

18	$\mathcal{F}_{\text{auth}}$	Authenticated transmission	C	[42, 25]
19	$\mathcal{F}_{\text{smt}}$	Secure message transmission	C	[118, 41]
20	$\mathcal{F}_{\text{sc}}$	Session secure channel	C	[61]
21	$\mathcal{F}_{\text{split}}$	Unauthenticated channels, split adversary	C	[16]
22	$\mathcal{F}_{\text{dauth}}$	On-line deniable authentication	C	[74]
23	$\mathcal{F}_{\text{diffuse}}$	Diffusion, gossip channel	C	[10]
24	$\mathcal{F}_{\text{rbc}}$	Reliable broadcast	C	[27]
25	$\mathcal{F}_{\text{BC}}$	Broadcast	C	[41, 93, 81]
26	$\mathcal{F}_{\text{BA}}$	Byzantine agreement	C	[111]
27	$\mathcal{F}_{\text{ABA}}$	Asynchronous Byzantine agreement	C	[65]
28	$\mathcal{F}_{\text{ledger}}$	Transaction ledger	C	[13]
29	$\mathcal{G}_{\text{ledger}}$	Global ledger, irreplaceability	C	[12]
30	$\mathcal{F}_{\text{certmail}}$	Certified mail	E	[117]

Layer 2. Cryptographic library and symmetric primitives

31	$\mathcal{F}_{\text{crypto}}$	Joint-state cryptographic library	C	[109, 106, 7, 56]
32	$\mathcal{F}_{\text{SE}}$	Symmetric encryption, stream cipher	C	[107, 7]
33	$\mathcal{F}_{\text{AE}}$	Authenticated encryption, on-line AE	C	[7]
34	$\mathcal{F}_{\text{fsAEAD}}$	Forward-secure AEAD	C	[59]
35	$\mathcal{F}_{\text{MAC}}$	Message authentication code	C	[7]
36	$\mathcal{F}_{\text{KDF}}$	Key derivation	C	[109]
37	$\mathcal{F}_{\text{OWF}}$	One-way function	O	—
38	$\mathcal{F}_{\text{PRG}}$	Pseudorandom generator	O	—
39	$\mathcal{F}_{\text{PRF}}, \mathcal{F}_{\text{PRP}}$	Pseudorandom function, permutation	E	[109]
40	$\mathcal{F}_{\text{CRHF}}$	Collision-resistant hashing	O	—
41	$\mathcal{F}_{\text{MHF}}$	Memory-hard function, password hashing	O	—

Layer 3. Public-key primitives, key exchange, messaging

42	$\mathcal{F}_{\text{PKE}}$	Public-key encryption, KEM	C	[41, 89, 106]
43	$\mathcal{F}_{\text{rPKE}}$	Replayable CCA relaxation	C	[62]
44	$\mathcal{F}_{\text{NCE}}$	Non-committing encryption	C	[48, 24, 115]
45	$\mathcal{F}_{\text{aPKE}}$	Adaptively secure non-interactive PKE	C	[53]
46	$\mathcal{F}_{\text{SIG}}$	Digital signature	C	[42, 6, 106]
47	$\mathcal{F}_{\text{adsig}}$	Adaptor signature	C	[85]
48	$\mathcal{F}_{\text{KE}}$	Key exchange, authenticated KE	C	[60, 26]
49	$\mathcal{F}_{\text{DH}}$	Ideal Diffie-Hellman exponentiation	C	[105]

continued on next page

#	Functionality	Reading	St.	References
50	$\mathcal{F}_{\text{pwKE}}$	Password-authenticated KE	C	[54, 86]
51	$\mathcal{F}_{\text{aPAKE}}$	Asymmetric PAKE	C	[98]
52	$\mathcal{F}_{\text{saPAKE}}$	Strong asymmetric PAKE	C	[98]
53	$\mathcal{F}_{\text{PHE}}$	Password-hardened encryption, threshold, key rotation	C	[1]
54	$\mathcal{F}_{\text{CKE}}$	Continuous KE, asymmetric ratchet	C	[59]
55	$\mathcal{F}_{\text{secmmsg}}$	End-to-end secure messaging	C	[59]
56	$\mathcal{F}_{\text{CGKA}}$	Continuous group key agreement	E	[5]†
57	$\mathcal{F}_{\text{IBE}}$	Identity-based encryption	E	—
58	$\mathcal{F}_{\text{ABE}}$	Attribute-based encryption	E	—
59	$\mathcal{F}_{\text{FE}}$	Functional, predicate encryption	O	[34]
60	$\mathcal{F}_{\text{FHE}}$	Threshold FHE decryption	C	[129]
61	$\mathcal{F}_{\text{obf}}$	Obfuscation, split VBB, iO	O	[18, 52]
Layer 4. Commitments and proofs				
62	$\mathcal{F}_{\text{COM}}$	Commitment	C	[49]
63	$\mathcal{F}_{\text{MCOM}}$	Multi-session commitment	C	[49]
64	$\mathcal{F}_{\text{NMCOM}}$	Non-malleable commitment	C	[70, 79]
65	$\mathcal{F}_{\text{eqv}}$	Equivocal, adaptively secure commitment	C	[23]
66	$\mathcal{F}_{\text{CP}}$	Commit-and-prove	C	[64]
67	$\mathcal{F}_{\text{ZK}}$	Zero knowledge	C	[64, 90]
68	$\mathcal{F}_{\text{NIZK}}$	Non-interactive zero knowledge	C	[91, 121]
69	$\mathcal{F}_{\text{SNARK}}$	Witness-succinct UC NIZK	C	[80, 69, 102]
70	$\mathcal{F}_{\text{acc}}$	Accumulator, vector commitment	C	[14]
Layer 5. Oblivious transfer and correlated randomness				
71	$\mathcal{F}_{\text{OT}}$	Oblivious transfer	C	[120, 77, 82, 116]
72	$\mathcal{F}_{\text{COT}}, \mathcal{F}_{\text{ROT}}$	Correlated, random OT	C	[101]†
73	$\mathcal{F}_{\text{OLE}}$	Oblivious linear evaluation	E	[35]†
74	$\mathcal{F}_{\text{OPRF}}$	Oblivious PRF	C	[97]†
75	$\mathcal{F}_{\text{GC}}$	Garbled circuit	E	—
76	$\mathcal{F}_{\text{PSI}}$	Private set intersection	E	—
77	$\mathcal{F}_{\text{PIR}}$	Private information retrieval	O	—
78	$\mathcal{F}_{\text{ORAM}}$	Oblivious RAM	O	—
Layer 6. Secret sharing, threshold cryptography, MPC				
79	$\mathcal{F}_{\text{VSS}}$	Verifiable secret sharing	C	[2, 84]
80	$\mathcal{F}_{\text{DKG}}$	Distributed key generation	C	[124]
81	$\mathcal{F}_{\text{thdec}}$	Threshold decryption	C	[2, 71]
82	$\mathcal{F}_{\text{TSIG}}$	Threshold signature	C	[51]
83	$\mathcal{F}_{\text{ABB}}$	Arithmetic black box	C	[71]†
84	$\mathcal{F}_{\text{SFE}}$	Two-party secure function evaluation	C	[126, 87]
85	$\mathcal{F}_{\text{MPC}}$	n -party secure computation	C	[64, 28]
86	$\mathcal{F}_{\text{fairSFE}}$	Guaranteed output delivery	C	[88, 92]
87	$\mathcal{F}_{\text{asyncMPC}}$	Asynchronous MPC	C	[27, 29]
88	$\mathcal{F}_{\text{incoerc}}$	Incoercible, receipt-free computation	C	[50]
89	$\mathcal{F}_{\text{coin}}$	Coin tossing	C	[32]
90	$\mathcal{F}_{\text{beacon}}$	Randomness beacon	C	[21]†
Layer 7. Privacy and anonymity				
91	$\mathcal{F}_{\text{mix}}$	Mix-net, verifiable shuffle	C	[123]
92	$\mathcal{F}_{\text{gsig}}$	Group signature	E	[123]
93	$\mathcal{F}_{\text{bsig}}$	Blind signature	C	[78, 3]
94	$\mathcal{F}_{\text{rsig}}$	Ring signature	C	[128]†, [36]

continued on next page

#	Functionality	Reading	St.	References
95	$\mathcal{F}_{\text{cred}}$	Anonymous credential, DAA	C	[39]
96	$\mathcal{F}_{\text{onion}}$	Onion routing	C	[40]
97	$\mathcal{F}_{\text{VRF}}$	Verifiable random function	C	[73]
Layer 8. Time and application composites				
98	$\mathcal{F}_{\text{TLP}}$	Time-lock puzzle, verifiable delay function	C	[22, 122]
99	$\mathcal{F}_{\text{chan}}$	State channel, virtual channel	C	[76, 75]
100	$\mathcal{F}_{\text{vote}}$	Self-tallying election	C	[4]

3 Modifier index

Each modifier is a transformation on functionalities rather than a functionality in its own right. Applied to the index above, the seven entries below generate several hundred distinct specifications, which is why they belong in a separate table.

Modifier	Effect on $\mathcal{F}_X$	References
SPLIT	Replicates $\mathcal{F}_X$ across adversarial network partitions; no authenticated setup	[16]
LEAKY	Adds a leakage interface exposing a bounded function of honest state	[31, 114]
INCOERCIBLE	Adds a fake-transcript interface so that inputs cannot be proved after the fact	[50]
DENIABLE	Removes simulator-side evidence of participation	[74]
FAIR	Removes adversarial abort; guarantees output delivery	[88, 92]
ADAPTIVE	Admits post-hoc corruption with state disclosure	[48, 23]
REPLAYABLE	Weakens the security relation, for example CCA to rCCA	[62]
JOINT STATE	Lets many sessions share one instance of $\mathcal{F}_X$, with local computation	[66, 106]
SID-FREE	Removes the pre-established session identifier that joint-state composition normally assumes	[108, 19]

4 Notes on coverage

Where the open entries cluster. The O markers fall almost entirely in Layer 2 and in the functional encryption band of Layer 3. This is structural rather than accidental. Layer 2 consists of local computations with no inter-party interface. The functional encryption band concerns what a key holder learns from a ciphertext, and simulation-based security there is impossible in general for unbounded key queries [34]. Entry 42 sits on the boundary between the two: the security statement for a memory-hard function is a cost lower bound rather than an indistinguishability, so it resists UC formulation for a third reason again.

Realizability boundaries worth recording alongside each entry. $\mathcal{F}_{\text{COM}}$, $\mathcal{F}_{\text{ZK}}$, $\mathcal{F}_{\text{OT}}$ and most of $\mathcal{F}_{\text{SFE}}$ are unrealizable in the plain model [49, 63]. $\mathcal{F}_{\text{BC}}$ as usually written is not realizable against adaptive adversaries, and must be weakened [93, 81]. $\mathcal{F}_{\text{coin}}$ with guaranteed output is unrealizable for

two parties, so the abort variant is the operative one. $\mathcal{F}_{\text{smt}}$ under adaptive corruption requires non-committing encryption [48], which is the same commitment problem that blocks naive symmetric-encryption functionalities.

Coverage sweep, 2022 to 2026. Recent work was located by web search and by querying the IACR ePrint archive; Google Scholar blocks automated queries, so no claim of exhaustiveness is made here, and the sweep is biased toward papers whose titles announce composability. It changed the table in five places rather than merely adding rows. Threshold FHE decryption [129] moves $\mathcal{F}_{\text{FHE}}$ from open to canonical, retiring one of the longest-standing gaps. Password-hardened encryption [1] enters as a new category adjacent to $\mathcal{F}_{\text{MHF}}$: it is the one recent functionality that touches password hashing, and the same paper reports a flaw in the proof of an earlier threshold construction. Adaptor signatures [85] enter as $\mathcal{F}_{\text{ad sig}}$. Ring signatures [128, 36] and self-tallying elections [4] acquire anchors they previously lacked. Most consequentially, the global restricted observable generic group of [33] now anchors $\mathcal{G}_{\text{GG}}$, which had no reference at all: it stands to the plain generic group exactly as $\mathcal{G}_{\text{RO}}$ stands to $\mathcal{F}_{\text{RO}}$, and it establishes UC security of a deployed proof system without asymptotic overhead. The complementary result that some deployed zkSNARKs are unconditionally UC-secure in the random oracle model [69] strengthens $\mathcal{F}_{\text{SNARK}}$.

Two merges preserved the count. The bilinear group models fold into $\mathcal{G}_{\text{GG}}$, since a bilinear group is a generic group with a pairing and the observable global functionality generalizes both. Verifiable delay functions fold into $\mathcal{F}_{\text{TLP}}$. Both follow the practice already used for $\mathcal{F}_{\text{PRF}}$ with $\mathcal{F}_{\text{PRP}}$ and for $\mathcal{F}_{\text{COT}}$ with $\mathcal{F}_{\text{ROT}}$. The merges are lossy in one respect worth recording: a verifiable delay function is publicly verifiable and a time-lock puzzle need not be, so the row understates a real distinction.

Deliberate exclusions. Witness encryption, aggregate and multi-signatures, fair exchange, electronic cash, cross-chain bridges, and composable analyses of operating-system services [44] were cut to hold the count at 100. Each is a reasonable candidate; none has a settled functionality that would displace an entry above.

Two source bibliographies. Entries are anchored against two published reference lists: that of the UC framework paper [41, 43], and that of the IITM model paper [110, 103]. The second contributes one new functionality, ideal Diffie-Hellman exponentiation [105], which is the realizable counterpart to assuming a generic-group setup and therefore now anchors $\mathcal{G}_{\text{GG}}$ as well as $\mathcal{F}_{\text{DH}}$. It also supplies the joint-state theorems for encryption and signatures with local computation [106] and their session-identifier-free refinement [108], both of which enter the modifier index rather than the main table. Where the two lists overlap they agree on page and volume data for seven entries, which are marked `status:cross-checked` in the companion `.bib` file. They diverge once: the UC list cites Canetti and Herzog’s J. Cryptology 24(1) article, the IITM list a distinct TCC 2006 paper [55]. Both are retained.

Two entries were displaced to hold the count at 100. Quantum and classical composability [30] is a framework variant rather than a functionality and moves to the paragraph above. Witness encryption remains excluded for the reason given earlier.

Defects in the source reference list. Four issues in the attached UC bibliography affect anyone mining it for citation keys. The entry keyed [b01] has an empty author surname; the author is Barak [15]. The key [y82] is used twice, for *Theory and Applications of Trapdoor Functions* and for *Protocols for Secure Computation*, both FOCS 1982; they are split here into y82a and y82b. The entry [bpw07] appears twice verbatim across a page break. Finally, [cdpw07] is titled *Universally Composable Security with Pre-Existing Setup*, whereas the published TCC 2007 title reads *with Global Setup*; a title variant rather than an error, but it will not match a DBLP lookup.

Framework variants and further limitations. Retained in the bibliography without a table entry, since they bear on how any of the above is stated rather than on what is stated: joint-state

composition [66, 106] and its session-identifier-free refinement [108, 19], global subroutines [9], the simplified variant for standard MPC [46], the IITM model [103, 110], GNUC [96] and SUC [125], RSIM [8], task-PIOAs [45], abstract and constructive cryptography [113, 112], local adversaries [68], quantum and classical composability [30], the runtime notions [95], the comparison of simulatability notions [104], the realizability obstruction for game-derived functionalities [72], the incompleteness of oblivious transfer in a synchronous model [94], the trusted-setup-free relaxations [119, 20], and symbolic analysis [55]. Trapdoor functions [127] are listed to resolve the key collision noted above.

References

- [1] Behzad Abdolmaleki, Ruben Baecker, Paul Gerhart, Mike Graf, Mojtaba Khalili, Daniel Rausch, and Dominique Schröder. Universally composable password-hardened encryption. *Cryptology ePrint Archive*, Paper 2025/1647, 2025.
- [2] Masayuki Abe and Serge Fehr. Adaptively secure feldman VSS and applications to universally-composable threshold cryptography. In *Advances in Cryptology – CRYPTO 2004*, pages 317–334, 2004.
- [3] Masayuki Abe and Miyako Ohkubo. A framework for universally composable non-committing blind signatures. *International Journal of Applied Cryptography*, 2(3):229–249, 2012.
- [4] Léo Ackermann, Myrto Arapinis, Pavlos Georgiou, Nikolaos Lamprou, Lenka Marešková, and Thomas Zacharias. E-cclasia: Universally composable self-tallying elections over anonymous broadcast. *IACR Communications in Cryptology*, 2(3), 2025.
- [5] Joël Alwen, Sandro Coretti, Yevgeniy Dodis, and Yiannis Tselekounis. Security analysis and improvements for the IETF MLS standard for group messaging. In *Advances in Cryptology – CRYPTO 2020*, 2020.
- [6] Michael Backes and Dennis Hofheinz. How to break and repair a universally composable signature functionality. In *Information Security Conference (ISC)*, 2004.
- [7] Michael Backes, Birgit Pfitzmann, and Michael Waidner. A composable cryptographic library with nested operations. In *10th ACM Conference on Computer and Communications Security (CCS)*, 2003. Extended version: ePrint 2003/015.
- [8] Michael Backes, Birgit Pfitzmann, and Michael Waidner. The reactive simulatability (RSIM) framework for asynchronous systems. *Information and Computation*, 205(12):1685–1720, 2007.
- [9] Christian Badertscher, Ran Canetti, Julia Hesse, Björn Tackmann, and Vassilis Zikas. Universal composition with global subroutines: Capturing global setup within plain UC. In *18th Theory of Cryptography Conference (TCC), Part III*, volume 12552 of *LNCS*, pages 1–30, 2020.
- [10] Christian Badertscher, Peter Gaži, Aggelos Kiayias, Alexander Russell, and Vassilis Zikas. Ouroboros genesis: Composable proof-of-stake blockchains with dynamic availability. In *ACM Conference on Computer and Communications Security (CCS)*, pages 913–930, 2018.
- [11] Christian Badertscher, Peter Gaži, Aggelos Kiayias, Alexander Russell, and Vassilis Zikas. Dynamic ad hoc clock synchronization. In *Advances in Cryptology – EUROCRYPT 2021*, volume 12698 of *LNCS*, pages 399–428, 2021.
- [12] Christian Badertscher, Julia Hesse, and Vassilis Zikas. On the (ir)replaceability of global setups, or how (not) to use a global ledger. In *19th Theory of Cryptography Conference (TCC), Part II*, 2021. ePrint 2020/1489.
- [13] Christian Badertscher, Ueli Maurer, Daniel Tschudi, and Vassilis Zikas. Bitcoin as a transaction ledger: A composable treatment. In *Advances in Cryptology – CRYPTO 2017*, 2017.

- [14] Foteini Baldimtsi, Ran Canetti, and Sophia Yakoubov. Universally composable accumulators. In *Topics in Cryptology – CT-RSA 2020*, volume 12006 of *LNCS*, pages 638–666, 2020. ePrint 2018/1241. Proves equivalence with the game-based accumulator definitions.
- [15] Boaz Barak. How to go beyond the black-box simulation barrier. In *42nd IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 106–115, 2001.
- [16] Boaz Barak, Ran Canetti, Yehuda Lindell, Rafael Pass, and Tal Rabin. Secure computation without authentication. *Journal of Cryptology*, 24(4):720–760, 2011.
- [17] Boaz Barak, Ran Canetti, Jesper Buus Nielsen, and Rafael Pass. Universally composable protocols with relaxed set-up assumptions. In *45th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 186–195, 2004.
- [18] Boaz Barak, Oded Goldreich, Russell Impagliazzo, Steven Rudich, Amit Sahai, Salil P. Vadhan, and Ke Yang. On the (im)possibility of obfuscating programs. In *Advances in Cryptology – CRYPTO 2001*, 2001. Journal version: J. ACM 59(2), 2012.
- [19] Boaz Barak, Yehuda Lindell, and Tal Rabin. Protocol initialization for the framework of universal composability. Cryptology ePrint Archive, Report 2004/006, 2004.
- [20] Boaz Barak and Amit Sahai. How to play almost any mental game over the net: Concurrent composition via super-polynomial simulation. In *46th IEEE Symposium on Foundations of Computer Science (FOCS)*, 2005.
- [21] Carsten Baum, Bernardo David, Rafael Dowsley, Ravi Kishore, Jesper Buus Nielsen, and Sabine Oechsner. CRAFT: Composable randomness beacons and output-independent abort MPC from time. In *Public-Key Cryptography (PKC)*, 2023.
- [22] Carsten Baum, Bernardo David, Rafael Dowsley, Jesper Buus Nielsen, and Sabine Oechsner. TARDIS: A foundation of time-lock puzzles in UC. In *Advances in Cryptology – EUROCRYPT 2021, Part III*, volume 12698 of *LNCS*, pages 429–459, 2021.
- [23] Donald Beaver. Adaptive zero-knowledge and computational equivocation. In *28th ACM Symposium on Theory of Computing (STOC)*, 1996.
- [24] Donald Beaver. Plug and play encryption. In *Advances in Cryptology – CRYPTO 1997*, 1997.
- [25] Mihir Bellare, Ran Canetti, and Hugo Krawczyk. A modular approach to the design and analysis of authentication and key exchange protocols. In *30th ACM Symposium on Theory of Computing (STOC)*, 1998.
- [26] Mihir Bellare and Phillip Rogaway. Entity authentication and key distribution. In *Advances in Cryptology – CRYPTO 1993*, volume 773 of *LNCS*, pages 232–249, 1994.
- [27] Michael Ben-Or, Ran Canetti, and Oded Goldreich. Asynchronous secure computation. In *25th ACM Symposium on Theory of Computing (STOC)*, pages 52–61, 1993.
- [28] Michael Ben-Or, Shafi Goldwasser, and Avi Wigderson. Completeness theorems for non-cryptographic fault-tolerant distributed computation. In *20th ACM Symposium on Theory of Computing (STOC)*, pages 1–10, 1988.
- [29] Michael Ben-Or, Boaz Kelmer, and Tal Rabin. Asynchronous secure computations with optimal resilience. In *13th ACM Symposium on Principles of Distributed Computing (PODC)*, pages 183–192, 1994.
- [30] Michael Ben-Or and Dominic Mayers. General security definition and composability for quantum and classical protocols. arXiv:quant-ph/0409062, 2004.

- [31] Nir Bitansky, Ran Canetti, and Shai Halevi. Leakage-tolerant interactive protocols. In *9th Theory of Cryptography Conference (TCC)*, pages 266–284, 2012.
- [32] Manuel Blum. Coin flipping by telephone. In *IEEE Spring COMPCON*, pages 133–137, 1982.
- [33] Jan Bobolz, Pooya Farshim, Markulf Kohlweiss, and Akira Takahashi. The brave new world of global generic groups and UC-secure zero-overhead SNARKs. In *22nd Theory of Cryptography Conference (TCC), Part I*, volume 15364 of *LNCS*, pages 90–124, 2024. ePrint 2024/818.
- [34] Dan Boneh, Amit Sahai, and Brent Waters. Functional encryption: Definitions and challenges. In *8th Theory of Cryptography Conference (TCC)*, 2011.
- [35] Elette Boyle, Geoffroy Couteau, Niv Gilboa, and Yuval Ishai. Compressing vector OLE. In *ACM Conference on Computer and Communications Security (CCS)*, 2018.
- [36] Pedro Branco, Nico Döttling, and Stella Wöhrig. Universal ring signatures in the standard model. In *Advances in Cryptology – ASIACRYPT 2022, Part IV*, volume 13794 of *LNCS*, pages 249–278, 2022.
- [37] Christina Brzuska, Marc Fischlin, Heike Schröder, and Stefan Katzenbeisser. Physically uncloneable functions in the universal composition framework. In *Advances in Cryptology – CRYPTO 2011*, 2011.
- [38] Jan Camenisch, Manu Drijvers, Tommaso Gagliardoni, Anja Lehmann, and Gregory Neven. The wonderful world of global random oracles. In *Advances in Cryptology – EUROCRYPT 2018*, 2018.
- [39] Jan Camenisch, Manu Drijvers, and Anja Lehmann. Universally composable direct anonymous attestation. In *Public-Key Cryptography (PKC)*, 2016.
- [40] Jan Camenisch and Anna Lysyanskaya. A formal treatment of onion routing. In *Advances in Cryptology – CRYPTO 2005*, 2005.
- [41] Ran Canetti. Universally composable security: A new paradigm for cryptographic protocols. In *42nd IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 136–145, 2001. Journal version: J. ACM 67(5), 2020.
- [42] Ran Canetti. Universally composable signature, certification, and authentication. In *17th IEEE Computer Security Foundations Workshop (CSFW)*, pages 219–233, 2004. Long version: ePrint 2003/239.
- [43] Ran Canetti. Universally composable security. *Journal of the ACM*, 67(5):1–94, 2020.
- [44] Ran Canetti, Suresh Chari, Shai Halevi, Birgit Pfitzmann, Arnab Roy, Michael Steiner, and Wietse Venema. Composable security analysis of OS services. In *Applied Cryptography and Network Security (ACNS)*, pages 431–448, 2011.
- [45] Ran Canetti, Ling Cheung, Dilsun Kaynar, Moses Liskov, Nancy Lynch, Olivier Pereira, and Roberto Segala. Time-bounded task-PIOAs: A framework for analyzing security protocols. In *20th International Symposium on Distributed Computing (DISC)*, pages 238–253, 2006.
- [46] Ran Canetti, Asaf Cohen, and Yehuda Lindell. A simpler variant of universally composable security for standard multiparty computation. In *Advances in Cryptology – CRYPTO 2015, Part II*, volume 9216 of *LNCS*, pages 3–22, 2015. See also ePrint 2014/553.
- [47] Ran Canetti, Yevgeniy Dodis, Rafael Pass, and Shabsi Walfish. Universally composable security with global setup. In *4th Theory of Cryptography Conference (TCC)*, volume 4392 of *LNCS*, pages 61–85, 2007.

- [48] Ran Canetti, Uriel Feige, Oded Goldreich, and Moni Naor. Adaptively secure multi-party computation. In *28th ACM Symposium on Theory of Computing (STOC)*, 1996.
- [49] Ran Canetti and Marc Fischlin. Universally composable commitments. In *Advances in Cryptology – CRYPTO 2001*, 2001.
- [50] Ran Canetti and Rosario Gennaro. Incoercible multiparty computation. In *37th IEEE Symposium on Foundations of Computer Science (FOCS)*, 1996.
- [51] Ran Canetti, Rosario Gennaro, Steven Goldfeder, Nikolaos Makriyannis, and Udi Peled. UC non-interactive, proactive, threshold ECDSA with identifiable aborts. In *ACM Conference on Computer and Communications Security (CCS)*, 2020.
- [52] Ran Canetti, Shafi Goldwasser, and Oxana Poburinnaya. Adaptively secure two-party computation from indistinguishability obfuscation. In *12th Theory of Cryptography Conference (TCC), Part II*, pages 557–585, 2015. See also ePrint 2014/845.
- [53] Ran Canetti, Shai Halevi, and Jonathan Katz. Adaptively secure non-interactive public-key encryption. In *2nd Theory of Cryptography Conference (TCC)*, 2005.
- [54] Ran Canetti, Shai Halevi, Jonathan Katz, Yehuda Lindell, and Philip MacKenzie. Universally composable password-based key exchange. In *Advances in Cryptology – EUROCRYPT 2005*, volume 3494 of *LNCS*, pages 404–421, 2005.
- [55] Ran Canetti and Jonathan Herzog. Universally composable symbolic analysis of mutual authentication and key-exchange protocols. In *3rd Theory of Cryptography Conference (TCC)*, volume 3876 of *LNCS*, pages 380–403, 2006.
- [56] Ran Canetti and Jonathan Herzog. Universally composable symbolic analysis of cryptographic protocols. *Journal of Cryptology*, 24(1):83–147, 2011.
- [57] Ran Canetti, Kyle Hogan, Aanchal Malhotra, and Mayank Varia. Universally composable network time. In *IEEE Computer Security Foundations Symposium (CSF)*, 2017.
- [58] Ran Canetti, Abhishek Jain, and Alessandra Scafuro. Practical UC security with a global random oracle. In *ACM Conference on Computer and Communications Security (CCS)*, pages 597–608, 2014.
- [59] Ran Canetti, Palak Jain, Marika Swanberg, and Mayank Varia. Universally composable end-to-end secure messaging. In *Advances in Cryptology – CRYPTO 2022*, volume 13508 of *LNCS*, 2022. ePrint 2022/376. Decomposes Signal into $\mathcal{F}_{\text{SecMsg}}$, a continuous-key-exchange functionality, epoch-level symmetric ratchets, and a forward-secure AEAD functionality.
- [60] Ran Canetti and Hugo Krawczyk. Analysis of key-exchange protocols and their use for building secure channels. In *Advances in Cryptology – EUROCRYPT 2001*, volume 2045 of *LNCS*, pages 453–474, 2001. Extended version: ePrint 2001/040.
- [61] Ran Canetti and Hugo Krawczyk. Universally composable notions of key exchange and secure channels. In *Advances in Cryptology – EUROCRYPT 2002*, volume 2332 of *LNCS*, pages 337–351, 2002. Extended version: ePrint 2002/059.
- [62] Ran Canetti, Hugo Krawczyk, and Jesper Buus Nielsen. Relaxing chosen-ciphertext security. In *Advances in Cryptology – CRYPTO 2003*, 2003. ePrint 2003/174.
- [63] Ran Canetti, Eyal Kushilevitz, and Yehuda Lindell. On the limitations of universally composable two-party computation without set-up assumptions. *Journal of Cryptology*, 19(2):135–167, 2006.

- [64] Ran Canetti, Yehuda Lindell, Rafail Ostrovsky, and Amit Sahai. Universally composable two-party and multi-party secure computation. In *34th ACM Symposium on Theory of Computing (STOC)*, pages 494–503, 2002.
- [65] Ran Canetti and Tal Rabin. Optimal asynchronous byzantine agreement. In *25th ACM Symposium on Theory of Computing (STOC)*, pages 42–51, 1993.
- [66] Ran Canetti and Tal Rabin. Universal composition with joint state. In *Advances in Cryptology – CRYPTO 2003*, volume 2729 of *LNCS*, pages 265–281, 2003.
- [67] Ran Canetti, Daniel Shahaf, and Margarita Vald. Universally composable authentication and key-exchange with global PKI. In *Public-Key Cryptography (PKC), Part II*, volume 9615 of *LNCS*, pages 265–296, 2016. See also ePrint 2014/432.
- [68] Ran Canetti and Margarita Vald. Universally composable security with local adversaries. In *Security and Cryptography for Networks (SCN)*, pages 281–301, 2012. See also ePrint 2012/117.
- [69] Alessandro Chiesa and Giacomo Fenzi. zkSNARKs in the ROM with unconditional UC-security. In *22nd Theory of Cryptography Conference (TCC), Part I*, volume 15364 of *LNCS*, 2024. ePrint 2024/724.
- [70] Giovanni Di Crescenzo, Yuval Ishai, and Rafail Ostrovsky. Non-interactive and non-malleable commitment. In *30th ACM Symposium on Theory of Computing (STOC)*, pages 141–150, 1998.
- [71] Ivan Damgård and Jesper Buus Nielsen. Universally composable efficient multiparty computation from threshold homomorphic encryption. In *Advances in Cryptology – CRYPTO 2003*, 2003.
- [72] Anupam Datta, Ante Derek, John C. Mitchell, Ajith Ramanathan, and Andre Scedrov. Games and the impossibility of realizable ideal functionality. In *3rd Theory of Cryptography Conference (TCC)*, 2006.
- [73] Bernardo David, Peter Gaži, Aggelos Kiayias, and Alexander Russell. Ouroboros praos: An adaptively-secure, semi-synchronous proof-of-stake blockchain. In *Advances in Cryptology – EUROCRYPT 2018*, 2018.
- [74] Yevgeniy Dodis, Jonathan Katz, Adam D. Smith, and Shabsi Walfish. Composability and on-line deniability of authentication. In *6th Theory of Cryptography Conference (TCC)*, pages 146–162, 2009.
- [75] Stefan Dziembowski, Lisa Ekey, Sebastian Faust, Julia Hesse, and Kristina Hostáková. Multi-party virtual state channels. In *Advances in Cryptology – EUROCRYPT 2019*, volume 11476 of *LNCS*, pages 625–656, 2019.
- [76] Stefan Dziembowski, Sebastian Faust, and Kristina Hostáková. General state channel networks. In *ACM Conference on Computer and Communications Security (CCS)*, pages 949–966, 2018. ePrint 2018/320.
- [77] Shimon Even, Oded Goldreich, and Abraham Lempel. A randomized protocol for signing contracts. *Communications of the ACM*, 28(6):637–647, 1985.
- [78] Marc Fischlin. Round-optimal composable blind signatures in the common reference string model. In *Advances in Cryptology – CRYPTO 2006*, 2006.
- [79] Marc Fischlin and Roger Fischlin. Efficient non-malleable commitment schemes. In *Advances in Cryptology – CRYPTO 2000*, volume 1880 of *LNCS*, pages 413–428, 2000.

- [80] Chaya Ganesh, Yashvanth Kondi, Claudio Orlandi, Mahak Pancholi, Akira Takahashi, and Daniel Tschudi. Witness-succinct universally-composable SNARKs. In *Advances in Cryptology – EUROCRYPT 2023, Part II*, volume 14005 of *LNCS*, pages 315–346, 2023. ePrint 2022/1618.
- [81] Juan A. Garay, Jonathan Katz, Ranjit Kumaresan, and Hong-Sheng Zhou. Adaptively secure broadcast, revisited. In *30th ACM Symposium on Principles of Distributed Computing (PODC)*, 2011.
- [82] Juan A. Garay and Philip MacKenzie. Concurrent oblivious transfer. In *41st IEEE Symposium on Foundations of Computer Science (FOCS)*, 2000.
- [83] Rosario Gennaro, Anna Lysyanskaya, Tal Malkin, Silvio Micali, and Tal Rabin. Algorithmic tamper-proof (ATP) security: Theoretical foundations for security against hardware tampering. In *1st Theory of Cryptography Conference (TCC)*, volume 2951 of *LNCS*, 2004.
- [84] Rosario Gennaro, Michael O. Rabin, and Tal Rabin. Simplified VSS and fast-track multi-party computations with applications to threshold cryptography. In *17th ACM Symposium on Principles of Distributed Computing (PODC)*, pages 101–112, 1998.
- [85] Paul Gerhart, Daniel Rausch, and Dominique Schröder. Universally composable adaptor signatures. Cryptology ePrint Archive, Paper 2025/1363, 2025.
- [86] Oded Goldreich and Yehuda Lindell. Session-key generation using human passwords only. In *Advances in Cryptology – CRYPTO 2001*, 2001.
- [87] Oded Goldreich, Silvio Micali, and Avi Wigderson. How to play any mental game. In *19th ACM Symposium on Theory of Computing (STOC)*, pages 218–229, 1987.
- [88] Shafi Goldwasser and Leonid A. Levin. Fair computation of general functions in presence of immoral majority. In *Advances in Cryptology – CRYPTO 1990*, volume 537 of *LNCS*, 1990.
- [89] Shafi Goldwasser and Silvio Micali. Probabilistic encryption. *Journal of Computer and System Sciences*, 28(2):270–299, 1984.
- [90] Shafi Goldwasser, Silvio Micali, and Charles Rackoff. The knowledge complexity of interactive proof systems. *SIAM Journal on Computing*, 18(1):186–208, 1989.
- [91] Jens Groth, Rafail Ostrovsky, and Amit Sahai. New techniques for noninteractive zero-knowledge. *Journal of the ACM*, 59(3):11:1–11:35, 2012.
- [92] Martin Hirt and Ueli Maurer. Complete characterization of adversaries tolerable in secure multi-party computation. *Journal of Cryptology*, 13(1):31–60, 2000.
- [93] Martin Hirt and Vassilis Zikas. Adaptively secure broadcast. In *Advances in Cryptology – EUROCRYPT 2010*, 2010.
- [94] Dennis Hofheinz and Jörn Müller-Quade. A synchronous model for multi-party computation and the incompleteness of oblivious transfer. Cryptology ePrint Archive, Report 2004/016, 2004.
- [95] Dennis Hofheinz, Jörn Müller-Quade, and Dominique Unruh. Polynomial runtime in simulatability definitions. In *18th IEEE Computer Security Foundations Workshop (CSFW)*, pages 156–169, 2005.
- [96] Dennis Hofheinz and Victor Shoup. GNUC: A new universal composability framework. *Journal of Cryptology*, 28(3):423–508, 2015.
- [97] Stanislaw Jarecki, Aggelos Kiayias, and Hugo Krawczyk. Round-optimal password-protected secret sharing and T-PAKE in the password-only model. In *Advances in Cryptology – ASIACRYPT 2014*, 2014.

- [98] Stanislaw Jarecki, Hugo Krawczyk, and Jiayu Xu. OPAQUE: An asymmetric PAKE protocol secure against pre-computation attacks. In *Advances in Cryptology – EUROCRYPT 2018*, volume 10822 of *LNCS*, pages 456–486, 2018. ePrint 2018/163. Defines $\mathcal{F}_{\text{aPAKE}}$ and $\mathcal{F}_{\text{saPAKE}}$.
- [99] Jonathan Katz. Universally composable multi-party computation using tamper-proof hardware. In *Advances in Cryptology – EUROCRYPT 2007*, 2007.
- [100] Jonathan Katz, Ueli Maurer, Björn Tackmann, and Vassilis Zikas. Universally composable synchronous computation. In *10th Theory of Cryptography Conference (TCC)*, pages 477–498, 2013.
- [101] Marcel Keller, Emmanuela Orsini, and Peter Scholl. MASCOT: Faster malicious arithmetic secure computation with oblivious transfer. In *ACM Conference on Computer and Communications Security (CCS)*, 2016.
- [102] Ahmed E. Kosba, Zhichao Zhao, Andrew Miller, Yi Qian, T-H. Hubert Chan, Charalampos Papamanthou, Rafael Pass, Abhi Shelat, and Elaine Shi. C0C0: A framework for building composable zero-knowledge proofs. Cryptology ePrint Archive, Paper 2015/1093, 2015.
- [103] Ralf Küsters. Simulation-based security with inexhaustible interactive turing machines. In *19th IEEE Computer Security Foundations Workshop (CSFW)*, pages 309–320, 2006. Full and revised version: ePrint 2013/025.
- [104] Ralf Küsters, Anupam Datta, John C. Mitchell, and Ajith Ramanathan. On the relationships between notions of simulation-based security. *Journal of Cryptology*, 21(4):492–546, 2008.
- [105] Ralf Küsters and Daniel Rausch. A framework for universally composable diffie-hellman key exchange. In *38th IEEE Symposium on Security and Privacy (S&P)*, pages 881–900, 2017.
- [106] Ralf Küsters and Max Tuengerthal. Joint state theorems for public-key encryption and digital signature functionalities with local computation. In *21st IEEE Computer Security Foundations Symposium (CSF)*, pages 270–284, 2008. Full version: ePrint 2008/006.
- [107] Ralf Küsters and Max Tuengerthal. Universally composable symmetric encryption. In *IEEE Computer Security Foundations Symposium (CSF)*, pages 293–307, 2009.
- [108] Ralf Küsters and Max Tuengerthal. Composition theorems without pre-established session identifiers. In *18th ACM Conference on Computer and Communications Security (CCS)*, pages 41–50, 2011.
- [109] Ralf Küsters and Max Tuengerthal. Ideal key derivation and encryption in simulation-based security. In *Topics in Cryptology – CT-RSA 2011*, volume 6558 of *LNCS*, pages 161–179, 2011. ePrint 2010/295. Gives $\mathcal{F}_{\text{crypto}}$ covering public-key and symmetric encryption, MACs, signatures, key derivation, and nonce generation.
- [110] Ralf Küsters and Max Tuengerthal. The IITM model: A simple and expressive model for universal composability. *IACR Cryptology ePrint Archive*, 2013/025, 2013.
- [111] Yehuda Lindell, Anna Lysyanskaya, and Tal Rabin. On the composition of authenticated byzantine agreement. In *34th ACM Symposium on Theory of Computing (STOC)*, 2002.
- [112] Ueli Maurer. Constructive cryptography: A new paradigm for security definitions and proofs. In *Theory of Security and Applications (TOSCA)*, volume 6993 of *LNCS*, pages 33–56, 2011.
- [113] Ueli Maurer and Renato Renner. Abstract cryptography. In *Innovations in Computer Science (ICS)*, pages 1–21, 2011.
- [114] Silvio Micali and Leonid Reyzin. Physically observable cryptography. In *1st Theory of Cryptography Conference (TCC)*, volume 2951 of *LNCS*, 2004.

- [115] Jesper Buus Nielsen. Separating random oracle proofs from complexity theoretic proofs: The non-committing encryption case. In *Advances in Cryptology – CRYPTO 2002*, pages 111–126, 2002.
- [116] Chris Peikert, Vinod Vaikuntanathan, and Brent Waters. A framework for efficient and composable oblivious transfer. In *Advances in Cryptology – CRYPTO 2008*, 2008.
- [117] Birgit Pfitzmann, Matthias Schunter, and Michael Waidner. Provably secure certified mail. Technical Report RZ 3207 (#93253), IBM Research, Zurich, 2000.
- [118] Birgit Pfitzmann and Michael Waidner. A model for asynchronous reactive systems and its application to secure message transmission. In *IEEE Symposium on Security and Privacy*, pages 184–201, 2001. See also ePrint 2000/066.
- [119] Manoj Prabhakaran and Amit Sahai. New notions of security: Achieving universal composability without trusted setup. In *36th ACM Symposium on Theory of Computing (STOC)*, pages 242–251, 2004.
- [120] Michael O. Rabin. How to exchange secrets by oblivious transfer. Technical Report TR-81, Aiken Computation Laboratory, Harvard University, 1981.
- [121] Charles Rackoff and Daniel R. Simon. Non-interactive zero-knowledge proof of knowledge and chosen ciphertext attack. In *Advances in Cryptology – CRYPTO 1991*, 1991.
- [122] Ronald L. Rivest, Adi Shamir, and David A. Wagner. Time-lock puzzles and timed-release crypto. Technical Report MIT/LCS/TR-684, MIT, 1996.
- [123] Douglas Wikström. *On the Security of Mix-Nets and Hierarchical Group Signatures*. PhD thesis, KTH Royal Institute of Technology, 2005.
- [124] Douglas Wikström. Universally composable DKG with linear number of exponentiations. In *Security in Communication Networks (SCN)*, volume 3352 of *LNCS*, pages 263–277, 2005. SCN 2004 proceedings, published 2005.
- [125] Douglas Wikström. Simplified universal composability framework. In *13th Theory of Cryptography Conference (TCC), Part A1*, pages 566–595, 2016.
- [126] Andrew C. Yao. Protocols for secure computations. In *23rd IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 160–164, 1982.
- [127] Andrew C. Yao. Theory and applications of trapdoor functions. In *23rd IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 80–91, 1982.
- [128] Kazuki Yoneyama and Kazuo Ohta. Ring signatures: Universally composable definitions and constructions. *Information and Media Technologies*, 2(4):1038–1051, 2007.
- [129] Guy Zyskind, Doron Zarchy, Max Leibovich, and Chris Peikert. High-throughput universally composable threshold FHE decryption. In *ACM Conference on Computer and Communications Security (CCS)*, 2025. ePrint 2025/1781. CCS 2025 distinguished paper.